The slope formula

1

a 3 b -3 c -1 d -1

e $\frac{3}{5}$ f -1 g 0 h 0

2 No. Reversing the order does not affect the final outcome of the slope. As long as you maintain the correct order by subtracting the corresponding y and x coordinates, the slope should be the same.

3

a Undefined b Undefined c Defined d Defined

e Undefined f Defined g Undefined h Defined

4

a $y = 25$ b 5

5

a $t = 3$ b $t = 15$ c $d = 20$ d $c = -\frac{12}{7}$

e $t = -23$

6 890 m

7

a 0 b 4 c No

8

a undefined b No c 1

9

a Neither b Undefined c 0 d Neither

e Neither f Neither g 0

10

a The change in x is equal to zero. The line is vertical.

b The line is horizontal. The change in y is equal to zero.

11

a No

b Yes

c No

12 **D**

13

a 0

b Undefined

14 **B**

15 -20

Additional questions

16

a

i 14 ii 7 iii 2

b

i 12 ii 4 iii 3

c

i -12 ii 4 iii -3

d

i -6 ii 2 iii -3

17 Because a vertical line has no run. So the formula for the slope $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a zero denominator is undefined.

18 0

19

a 2 b $-\frac{7}{5}$ c $\frac{1}{2}$ d $-\frac{2}{3}$

20

a -2

b As x increases, the value of y decreases.

21

a 3

b $\frac{3}{5}$

c -4

d -3

e 0

f 0

g 1

h -1

22

a $\frac{5}{3}$

b $\frac{5}{3}$

c Yes

23 $m = 1$